Homework Assignment 5

You are free to choose to between parts 1 and 2 of this assignment. Both will earn you the same number of points. If you decide to submit more than one part, you will not receive extra credit but can make up for missed points. Part 1 is closer to the lab assignment. Part 2 is different and a bit more interesting.

Part 1 – Mandelbrot and Julia Sets

The Mandelbrot and Julia sets both rely on the very same complex iteration formula:

$$z_{n+1} = (z_n)^2 + C \quad . \quad (\#)$$

For Mandelbrot set, you always start with $z_0=0$ and C is the parameter that you vary. The Mandelbrot set is defined as set of all points C in the complex plane for which the series (#) does not diverge.

Conversely for the Julia set, the roles of z_0 and C are reversed. For this set, you vary the initial value, $z_0=x+y\times i$. The parameter C stays fixed at some complex number to be specified beforehand. For every C value, a somewhat different Julia set emerges. The Julia set is defined as the set of complex points z_0, for which the series (#) does not diverge.

- Start from your latest version of Mandelbrot program from this week's computer lab. Reverse the roles of C and z_0. Fix C at $-0.4+0.6\times i$. Change the initial value of z by setting $z_0=x+i\times y$. Investigate the interval $-3<x<+3$ and $-3<y<+3$ and locate the fractal region. Zoom in a little bit so that the fractal region fills the entire picture. Set your matrix to the iteration number n at which the series started to exceed z_{max}, as was explained in the lab handout. Generate an image in good resolution and submit it along with your notebook as a separate file.

- Now zoom in deeper and deeper again until the x and y interval is less than 0.001 in size. Insert your best image here and provide your x and y ranges. You may have to increase the maximum number of iterations. Please keep C fixed at $-0.4+0.6\times i$.

- Now zoom out again and change C to $-0.85+0.22\times i$. Generate a picture that shows the fractal region. Change the colors in your image by adding the new argument `cmap='inferno'` to the `imshow` command. Search online for available color maps and pick one you like.

- Now explore the C parameter region on your own. Zoom in until the interval size is less than 0.001. Submit your most beautiful picture as a separate file. Please specify c and the z_0 interval.

Part 2 – Newton Fractals

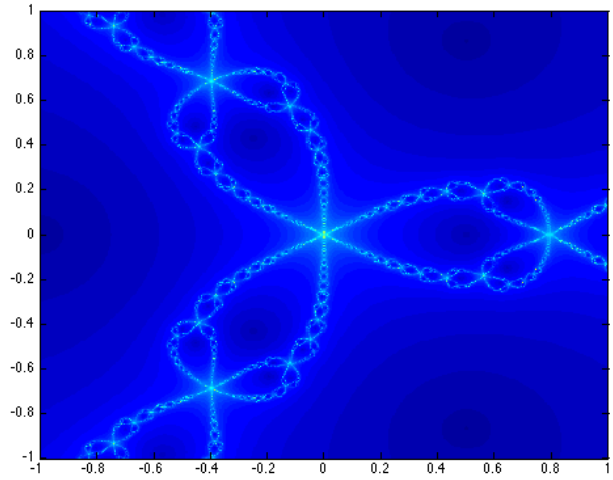
Newton's iterative method,

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)} \ ,$$

finds roots of equations, f(z)=0. It finds the zeros of a complex function f(z). For this assignment, please derive the specific iteration formula for the function $f(z)=z^3+1$,

$$z_{n+1} = z_n - \ldots$$

(1) Enter this formula into your Mandelbrot code and replace the iteration formula (#) with the equation above. There is no constant *C* in this problem. Use the coordinates x and y to define the initial point, $z_0=x+y\times i$, where the algorithm starts to look for zeros given by f(z)=0. We do not want to stop the iterations when $|z_n|$ gets very large but instead when the code it very close to an actual root. Use $|f(z_n)| < 10^{-3}$ to terminate the iterations. You should find an image like this:



(2) How many zeros does the function $f(z)=z^3+1$ have in the complex plane? Describe where they are located in the picture above. **Change your code so that it gives the pixel a different color depending on which of the three roots it converged to.**

(3) In original code from part (1), change the function to $f(z)=z^n+1$ for n=4, 5, and 6. **Submit the three resulting images** as separate image files in .png format with at least 200 dots per inch (dpi=200)**.**

(4) Finally apply Newton's method to the function $f(z) = (z-z_a) \times (z-z_b) \times (z-z_c) \times (z-z_d)$. The parameters z_a, z_b, z_c, and z_d are complex constants that you choose beforehand. Choose them so that an interesting picture emerges that is different from those that you generated in part (3). *(We are welcome to choose functional forms other than f(z) to see if new fascinating images emerge.)*